

Tidal locking as a negative feedback on Earth-like planetary dynamos: consequences for magnetic shielding and habitability

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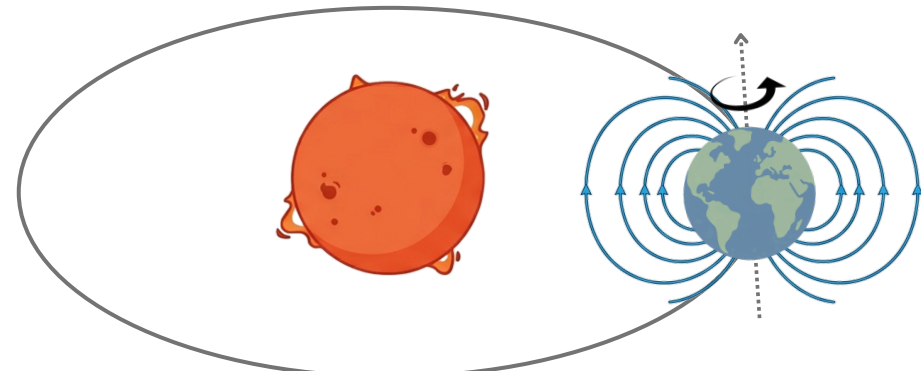
Context

- M dwarfs constitute the predominant stellar population in our galaxy, accounting for approximately 70–75% of all stars.
- In these systems, the **Habitable Zone (HZ)** is located significantly closer to the host star, due to their low luminosities L_* .

$$r_{\text{HZ}} = \left(\frac{L_*/L_\odot}{S_{\text{eff}}} \right)^{1/2} [\text{AU}] \quad \text{where } S_{\text{eff}} \text{ is the effective stellar flux.}$$

- Within these close-in HZs, terrestrial planets are highly susceptible to **tidal synchronization**, which increases their rotation period significantly.

- Can the intrinsic planetary magnetic shield survive the spin-down?**



Framework

* Planetary rotation rate

To ensure that rotation smoothly approaches synchronization, we solve the angular evolution equation from the classic Constant Time Lag (CTL) models (Barnes 2017). Assuming zero obliquity and zero eccentricity, it yields

$$\frac{d\omega}{dt} = \frac{3k_2}{\alpha M_{\text{pl}}} \frac{G^2 M_*^2 (M_* + M_{\text{pl}}) r_{\text{pl}}^3}{n r_{\text{orb}}^6} \tau \left(1 - \frac{\omega}{n} \right),$$

where k_2 is the second-degree Love number, α is the dimensionless moment of inertia, M_{pl} is the planetary mass, G is the gravitational constant, M_* is the mass of the host, n is the mean orbital motion, r_{orb} is the orbital distance, r_{pl} is the planetary radius, and τ is the tidal time lag. Assuming constant planetary parameters and a time-independent orbit, the solution is given by

$$\omega(t) = n + (\omega_i - n) \exp\left(-\frac{t}{\tau_{\text{lock}}}\right), \quad \text{with: } \tau_{\text{lock}} = \frac{1}{\tau} \frac{\alpha}{3k_2} \frac{M_{\text{pl}} r_{\text{orb}}^6}{G M_*^2 r_{\text{pl}}^3}$$

* Planetary dynamo scaling

For the planetary magnetic field, we assume a dipolar configuration, based on that of Earth, where $B_{\text{p},0}$ is the magnetic field at the pole.

$$B_{\text{pl}}(r) = \frac{1}{2} B_{\text{p},0} \left(\frac{r_{\text{pl}}}{r} \right)^3$$

To quantify how rotation affects the planetary magnetic field, we use the following models, based on dynamo scaling laws:

- Model A:** The planetary magnetic field scales directly with rotation.

$$B_{\text{p},0}^{(\omega)}(t) = (\omega(t)/\omega_i) B_{\text{p},0}$$

- Model B:** The magnetic field strength is primarily driven by the convective heat flux, and rotation plays a critical role in shaping the geometry of it (Christensen & Aubert, 2006).

A key diagnostic of the magnetic topologies of rotating dynamos, is the **local Rossby number**, where:

$$Ro_\ell(t) \approx Ro_{\ell,0} \left(\frac{\omega(t)}{\omega_i} \right)^{-7/6}$$

where:

$$Ro_{\ell,\text{crit}} = 0.12$$

$Ro_\ell < Ro_{\ell,\text{crit}}$

Dipolar field
 $f_{\text{dip}} = 1$

$Ro_\ell \geq Ro_{\ell,\text{crit}}$

Multipolar field
 $f_{\text{dip,TL}} \approx 0.1 f_{\text{dip}}$

$$B_{\text{p},0}^{(\omega)} = \left[f_{\text{dip,TL}} + \frac{f_{\text{dip}} - f_{\text{dip,TL}}}{1 + e^{k(Ro_\ell - Ro_{\ell,\text{crit}})}} \right] B_{\text{p},0}$$

* Magnetopause standoff distance

Finally, to estimate the size of the **upstream magnetosphere**:

$$\frac{r_{\text{M}}}{r_{\text{pl}}} = \sqrt[6]{\frac{f_0^2 B_{\text{p},0}^{(\omega)^2}}{2\mu_0 p_*(r_{\text{orb}})}}$$

where f_0 is the form factor, μ_0 is the magnetic permeability of vacuum, and p_* is the total stellar pressure.

The total pressure assumes an isotropic stellar wind, and it is based on empirical relations (Wright et al. 2018, Wood et al. 2021) and numerical simulations (e.g., Cranmer & Saar 2011).

Results

For the hypothetical planets, we model a **Modern Earth-like planet (ME)** with a rotation period of 24 hrs, an initial polar magnetic field of 1 G, and a tidal dissipation factor of $Q \approx 12$. Additionally, we model an **Early Earth-like planet (EE)**, based on the Paleoproterozoic, with an initial rotation period of 13 hrs, a weaker magnetic field of 0.7 G and $Q \approx 100$.

The sizes of the upstream magnetospheres of ME and EE planets, under models A and B, across the HZ of a sample of 15 M dwarfs from the literature are shown in Figure 1.

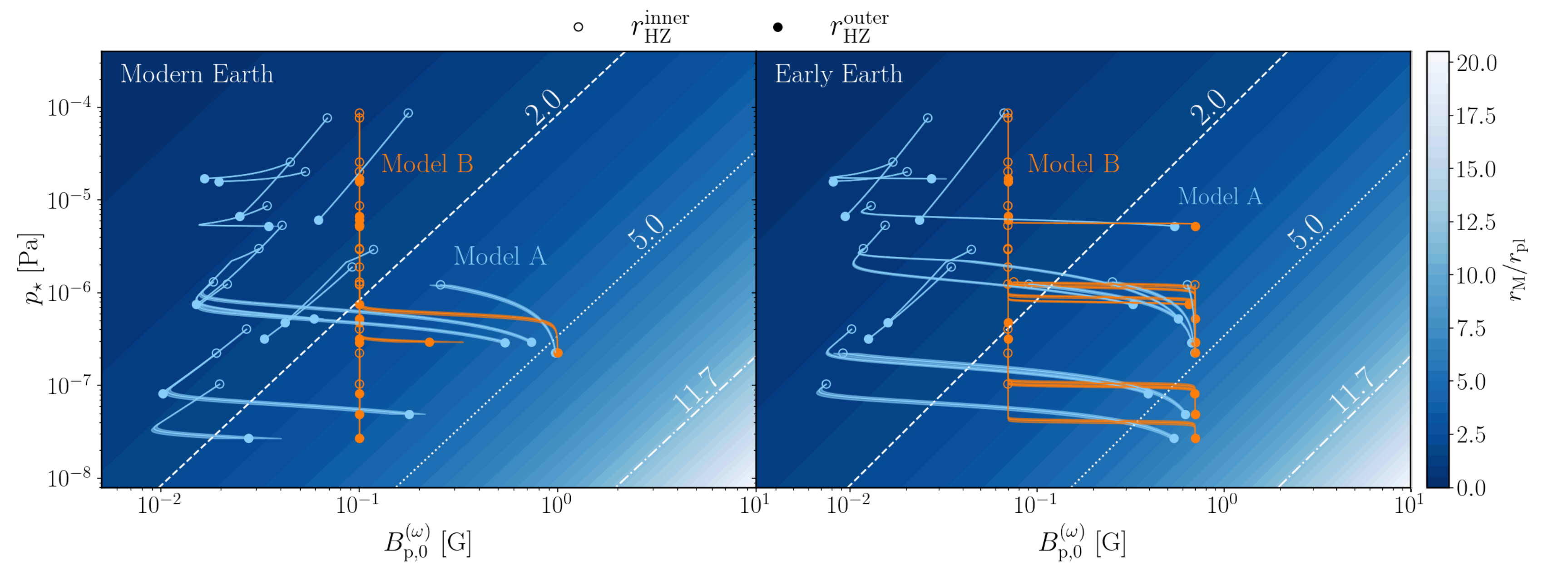


Figure 1. Normalized magnetospheres as a function of the planetary magnetic field and the total stellar pressure. The entire HZ of each star is displayed, where $\circ$ is the inner radius and $\bullet$ the outer radius.

Additionally, in Figure 2 the magnetospheric sizes as a function of the orbital distances of four representative cases are shown.

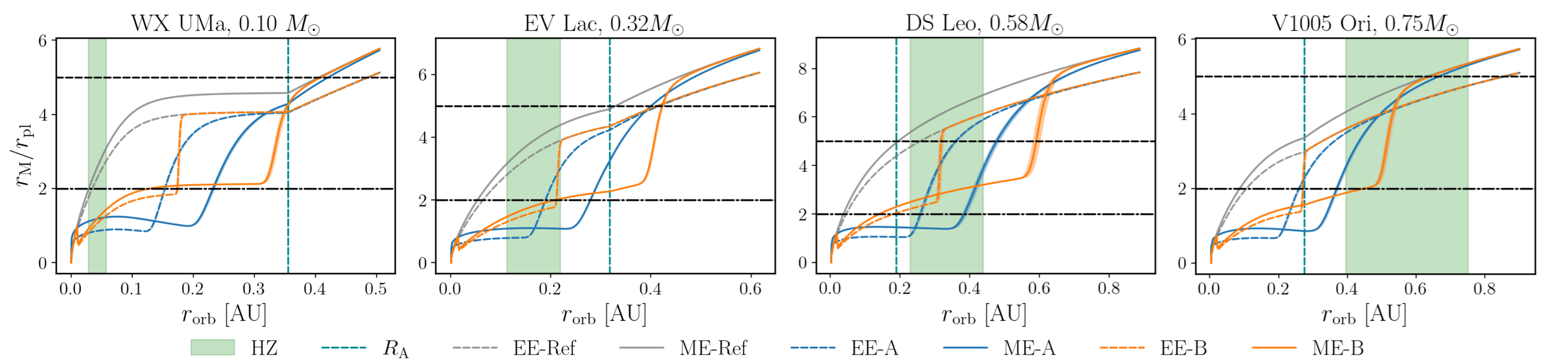


Figure 2. Magnetospheres of ME and EE, under models A and B as functions of the orbital distance from representative cases.

Following Rodríguez-Mozos & Moya (2019), we define an atmospheric protection likelihood $P(\alpha_0)$, where $r_{\text{M}}/r_{\text{pl}} \geq 5$ is assumed to be a likelihood of 1.0, and 0.0 is set at $r_{\text{M}}/r_{\text{pl}} \leq 2$ (Lammer et al. 2007). The likelihoods from the full sample are displayed in Figure 3.

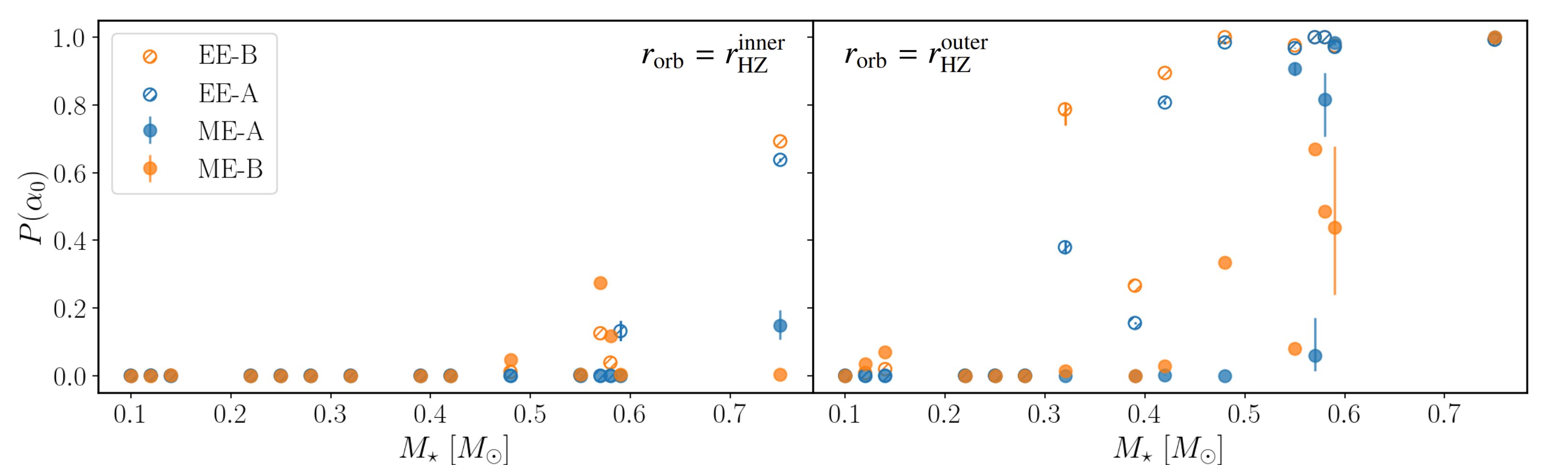


Figure 3. Likelihoods of atmospheric protection at the inner (left) and outer (right) boundaries of the HZ of our full sample.

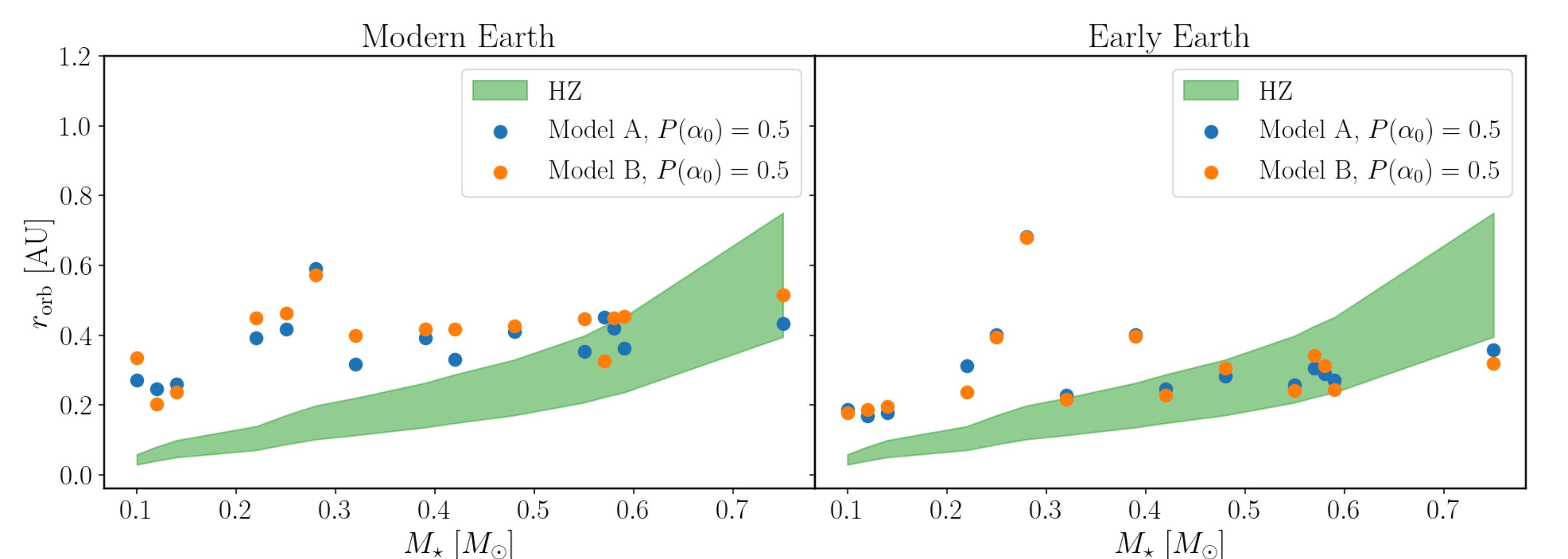


Figure 4. Orbital distance required to achieve an atmospheric protection likelihood of 0.5, as a function of mass.

Based on these atmospheric protection likelihoods, we conclude that:

- At the **inner HZ**, the likelihoods are **essentially zero** with all models across most of the sample.
- At the **outer HZ**, the situation is more favourable for early M dwarfs ($\geq 0.32 M_\odot$) hosting Early Earth.
- Tidal locking acts as a **severe negative feedback** on the planetary magnetic shielding.

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- PDF version
- Data used
- Model details

